

Distributed Cross-Learning for Equitable Federated models - privacy-preserving prediction on horizontal and vertical clinical data

Corresponding Author: Dr Tsung-Ting Kuo

This manuscript has been previously reviewed at another journal that is not operating a transparent peer review scheme. The manuscript was considered suitable for publication without further review at Nature Communications.

This file contains all reviewer reports in order by version, followed by all author rebuttals in order by version.

Version 0:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

The manuscript introduces Distributed Equitable Cross-Learning (DECL), a collaborative training algorithm designed to address security concerns inherent in Federated Learning systems. By eliminating the need for a third-party server to access locally trained models, DECL proposes a secure distributed training framework. This is achieved by maintaining a copy of the global model at each site, with secure updates facilitated through blockchain technology and distributed file system methods. Furthermore, the adoption of GLORE and VERTIGO allows DECL to accommodate both horizontally and vertically partitioned data, showcasing its versatility.

Major Areas for Improvement:

1. Justification for the Use of Distributed File Systems:

The authors propose utilizing a distributed file system to enhance the scalability of DECL for handling larger models. However, the empirical validation of this approach is conducted using a logistic regression model, which is inherently simple and linearly scales with the number of variables. Given the modest scale of the logistic regression model tested (involving 100 variables), the rationale for adopting a distributed file system—especially in an era dominated by large-scale models with billions or trillions of parameters—remains undermotivated. A more compelling justification, particularly how it scales with significantly larger and more complex models, would strengthen the case for this design choice.

2. Integration with Complex Models:

While employing logistic regression as a proof of concept for DECL is understandable, extending the framework to more complex models is essential for demonstrating its applicability in real-world scenarios. The manuscript would benefit from illustrating how DECL can be integrated with, and enhance the performance of, advanced modeling techniques beyond logistic regression. This expansion is critical for substantiating the algorithm's relevance and utility in diverse and sophisticated data analysis tasks.

3. Comprehensive Comparison with Existing Solutions:

The comparison of DECL to a centralized logistic regression model, while a starting point, does not suffice to illustrate its advantages comprehensively. Given the substantial dataset at hand, which includes over 15,000 patients and numerous covariates, there is a significant opportunity to benchmark DECL against state-of-the-art models, such as tree-based algorithms and deep neural networks.

(Remarks on code availability)

Reviewer #2

(Remarks to the Author)

This paper introduces the Distributed Equitable Cross-Learning (DECL) framework, which leverages the concepts of federated learning and blockchain to distribute models across horizontally or vertically partitioned data, aiming to protect patient privacy. The approach is evaluated on three different prediction tasks—COVID-19, total hip arthroplasty, and myocardial infarction—using real-world data from five University of California (UC) Health medical centers, UC San Diego, and Edinburgh, UK. The results show that the proposed method is promising, with prediction performance comparable to that of centralized modeling solutions. Overall, the paper is well-written and organized, but some issues need further attention, as outlined below:

- The characteristics of each dataset, including demographics and covariates, should be clearly outlined in a table.
- Incorporating and comparing contemporary federated learning methods with the DECL would highlight its strengths and weaknesses more effectively.
- The distribution of outcomes, especially the proportion of negative and positive samples in each dataset, needs clearer presentation to indicate whether the datasets are balanced.
- The evaluation should include a wider range of metrics, such as F1-score, recall, precision, and sensitivity, for a more detailed performance assessment beyond just AUC.
- While logistic regression is the primary model for both DECL and centralized learning, exploring other models like XGBoost could provide deeper insights, especially since logistic regression may not perform well in all scenarios. A variety of models should be tested for a thorough analysis.

(Remarks on code availability)

Version 2:

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Thanks for addressing our previous comments. In this version, the authors did incorporate their framework with slightly newer machine learning models, namely the XGboost and MLP. However, these two models are still the more conventional models. It will be a lot more relevant if the team can demonstrate the performance of their proposed framework with the recently popular models, such as transformers, CNNs and LSTM etc.

The concept of Federated Learning is not new. The study implemented an existing blockchain technology to support the model training and also a distributed file system for data storage. These ideas are not new neither. I fail to see the significant novelties and technical contributions from this study.

(Remarks on code availability)

Reviewer #2

(Remarks to the Author)

Thanks for addressing the comments comprehensively.

(Remarks on code availability)

Version 3:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

Thanks for addressing my previous comments. No further comments from me.

(Remarks on code availability)

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We agree that more justification for the use of distributed file systems will enhance our manuscript. Therefore, we included reference to more complex models, which need such a distributed file system for scalability, at the end of the [METHODS Section 2.4](#).

2. Integration with Complex Models: While employing logistic regression as a proof of concept for DECL is understandable, extending the framework to more complex models is essential for demonstrating its applicability in real-world scenarios. The manuscript would benefit from illustrating how DECL can be integrated with, and enhance the performance of, advanced modeling techniques beyond logistic regression. This expansion is critical for substantiating the algorithm's relevance and utility in diverse and sophisticated data analysis tasks.

Following this recommendation, we expanded our methods to integrate D-CLEF with Multi-Layer Perceptron (MLP) and eXtreme Gradient Boosting (XGBoost), by using ensemble between the privacy-preserving federated D-CLEF model and a siloed/locally constructed MLP/XGBoost model. This will allow a site, should they desire, to combine the benefits of a global federated model with a more complex local model. These changes are described in [METHODS Section 2.7](#). We also revised [Fig. 2–10](#) to reflect this change. We did not include tree-based models at this point given the scope of our current work, but will attend to this recommendation in future work, as mentioned in the [2nd Paragraph of DISCUSSION](#).

3. Comprehensive Comparison with Existing Solutions: The comparison of DECL to a centralized logistic regression model, while a starting point, does not suffice to illustrate its advantages comprehensively. Given the substantial dataset at hand, which includes over 15,000 patients and numerous covariates, there is a significant opportunity to benchmark DECL against state-of-the-art models, such as tree-based algorithms and deep neural networks.

We further implemented siloed and centralized MLP and XGBoost methods (as described in the revised [METHODS Section 2.8](#)) and compared them with D-CLEF as well as the ensemble with siloed MLP or XGBoost model. We expanded the number of our experiments from 2,490 to 6,150, as shown in the revised [METHODS Section 3](#). The evaluation process is described in the revised [SUPPLEMENTARY NOTES Section 2, 6 and 8](#). We updated the results in the [RESULTS Section 1–3](#), [SUPPLEMENTARY NOTES Section 3–4](#), [Fig. 2–10](#), as well as [Supplement Data 7–12, 15–20 and 23–28](#). In general, D-CLEF performs similar to the centralized models, while being better than siloed ones.

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- The characteristics of each dataset, including demographics and covariates, should be clearly outlined in a table.

We totally agree that inclusion of the demographics and covariates information is critical, and have added [Supplementary Data 3–6, 14, and 22](#) (for demographics), as well as [Supplementary Data 2, 13, and 21](#) (for covariates). Specifically, [Supplementary Data 2–6](#) are for dataset X, [Supplementary Data 13–14](#) are for dataset Y, and [Supplementary Data 21–22](#) are for dataset Z. Additionally, we added the discussion of the demographics of the excluded patients in dataset X in the [METHODS Section 1.1](#).

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We added the comparison to federated learning methods in the [5th Paragraph of INTRODUCTION](#), as well as adding a new [Supplementary Data 1](#), to demonstrate their major differences when compared with D-CLEF.

- The distribution of outcomes, especially the proportion of negative and positive samples in each dataset, needs clearer presentation to indicate whether the datasets are balanced.

We specifically describe the proportion and numbers of negative and positive samples in the [1st Paragraph of the RESULTS Section 1 – 3](#).

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- While logistic regression is the primary model for both DECL and centralized learning, exploring other models like XGBoost could provide deeper insights, especially since logistic regression may not perform well in all scenarios. A variety of models should be tested for a thorough analysis.

We agree with the reviewer's feedback, which is in line with Reviewer #1 – Bullet Point #2. In response to both reviewers, we implemented siloed and centralized Multi-Layer Perceptron (MLP) and eXtreme Gradient Boosting (XGBoost) methods (as described in the revised [METHODS Section 2.8](#)). Also, we expanded our method to integrate D-CLEF with a siloed MLP or XGBoost model via ensemble method (as described in the revised [METHODS Section 2.7](#)). We also revised [Fig. 2–10](#) to reflect this change. Then, we compared D-CLEF as well as its ensemble with siloed MLP/XGB with siloed/centralized Logistic Regression (LR), MLP and XGBoost models. We expanded the number of our experiments from 2,490 to 6,150, as shown in the revised [METHODS Section 3](#). The evaluation process is described in the revised [SUPPLEMENTARY NOTES Section 2, 6 and 8](#). We updated the results in the [RESULTS Section 1–3](#), [SUPPLEMENTARY NOTES Section 3–4](#), [Fig. 2–10](#), as well as [Supplement Data 7–12, 15–20 and 23–28](#). In general, D-CLEF's performance is close to the centralized models, while outperforming the siloed ones.

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Thank you very much!

However, these two models are still the more conventional models. It will be a lot more relevant if the team can demonstrate the performance of their proposed framework with the recently popular models, such as transformers, CNNs and LSTM etc.

We totally agree that demonstrating the performance of our proposed D-CLEF framework with the recently popular models is important. Therefore, we implemented and evaluated three new models: Two-Dimensional Convolutional Neural Network (CNN-2D), One-Dimensional Convolutional Neural Network (CNN-1D), and Long Short-Term Memory (LSTM), as described in the expanded [METHOD Section 2.7 and 3](#), as well as the [SUPPLEMENTARY NOTES Section 2](#). The experiment results showed that D-CLEF has the potential to improve its prediction capability by incorporating different types of models. We also updated [Fig. 2–10](#), as well as [Supplementary Data 7, 9, 10, 15, 17, 18, 23, 25, and 26](#), to include the new results. We also mentioned transformers as one of the future directions to explore in the [4th Paragraph](#) of the [DISCUSSION](#) section.

The concept of Federated Learning is not new. The study implemented an existing blockchain technology to support the model training and also a distributed file system for data storage. These ideas are not new neither. I fail to see the significant novelties and technical contributions from this study.

We appreciate the reminder of highlighting the novelty of our proposed method. Therefore, we summarized the technical novelties and contributions at the end of the [INTRODUCTION](#) section.

Reviewer #2 (Remarks to the Author):

Thanks for addressing the comments comprehensively.

Thank you very much for the positive feedback!

REVIEWER COMMENTS

Reviewer #1 (Remarks to the Author):

Thanks for addressing my previous comments. No further comments from me.

Thank you very much!